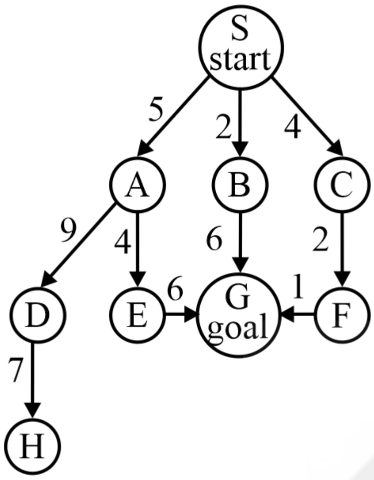
Artifical Intelligence

Miscellaneous

DPP: 1

Q1 Consider the state space as shown below:

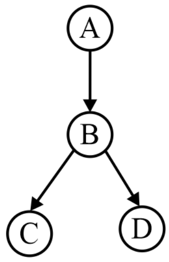


Apply BFS algorithm, and find out the path and cost of path using BFS.
Assume that nodes are generated in alphabetical order.
(A) Path: S->B->GCost: 8
(B) Path: S->A->E->GCost: 15
(C) Path: S->C->F->GCost: 7
(D) Search is incomplete, path not found

Q2 Two random variables, X and Y, have a joint probability distribution given by P(X = x,Y = y), The joint probabilities are as follows: P(X = 1,Y = 1) = 0.2, P(X = 1, Y = 2) = 0.3, P(X = 2, y = 1) = 0.4 and P(X = 2, Y = 2) = 0.1. What is the probability that Y = 1 given X = 1?
(A) 0,1 (B) 0,2
(C) 0,4 (D) 0,5

Q3 You have a bag of 20 marbles, 8 of which are red, and 12 are blue. You draw one marble from the bag, record its color (Event A), and then draw a second marble from the same bag without replacement and record its color (Event B). Calculate the probability that both marbles are red (Event A and Event B).

Q4 Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	P(·)
A = 0	0.6
A =1	0.4

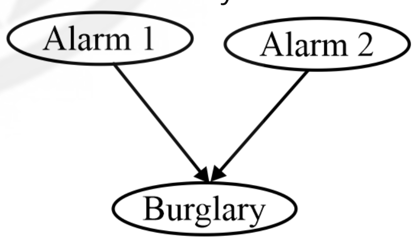
	P(B = 0|·)	**P(B = 1|·)**
A = 0	0.6	0.6
A = 1	0.4	0.4

	P(C = 0|·)	**P(C = 1|·)**
B = 0	1	0
B = 1	0	1

	P(D = 0|·)	**P(D = 1|·)**
B = 0	0.6	0.6
B = 1	1	1

The value of P (A = 1, B = 0, C = 0, D =1) ____.
(Rounded off to three decimal places)

Q5 Consider the Bayesian Network shown below:



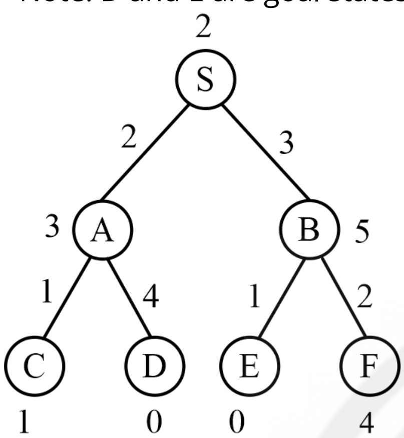
P(Alarm1) = 0.1
P(Alarm2) = 0.2
P(Burglary | Alarm1, Alarm2) = 0.8
P(Burglary | Alarm1, ¬ Alarm2) = 0.7
P(Burglary | Ø Alarm1, Alarm2) = 0.6
P(Burglary| Ø Alarm1, Ø Alarm2) = 0.5
Calculate P(Alarm2 | Burglary, Alarm1).
Enter ans upto two decimals

Q6 A Bayesian network consists of 4 binary variables. How many unique joint assignments can be generated using forward sampling from this network?

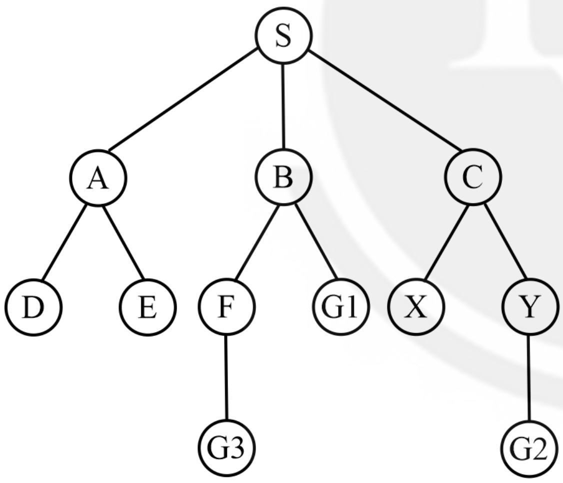
Q7 Evaluate the number of truth assignments for the proposition (P v Q) $\rightarrow$ R.

Q8 The optional path cost using A* algo for the root node is________.

Note: D and E are goal states.



Q9 Count the number of states expanded before reaching the goal state. Use alphabetical order for tree- break.

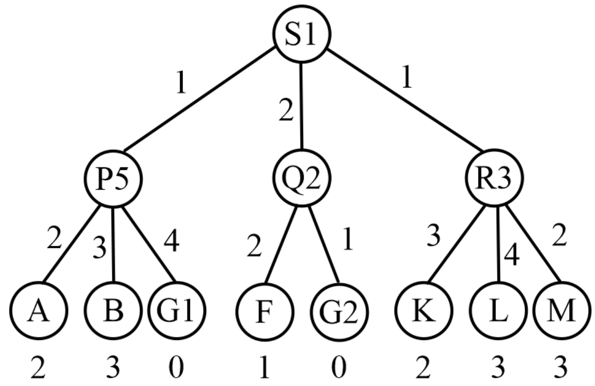


Q10 Simplify the following:

P Ù (P « Q) Ú Q

Q11 The optional path cost from root to goal nodes G1 and G2 using GB+S is____.

Use alphabetic order.



Q12 "x [x node 2 = 0, x Î {2x2 – 7x +3 = 0} Ç {x^2 + 4x + 4 = 0}]

S1: The truth value of above expression is true.

S2: The truth value of above expression is False.

The number of correct statement is__________.

Answer Key

Q1	A
Q2	C
Q3	$\frac{14}{95}$
Q4	0.096
Q5	0.22
Q6	16
Q7	5
Q8	4
Q9	6
Q10	00
Q11	3
Q12	1

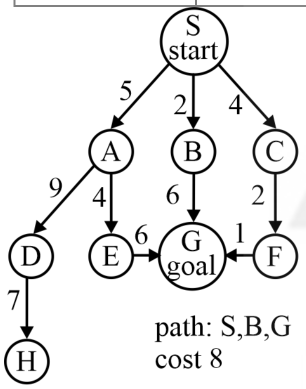
Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

# of nodes tested: 7, expanded: 6

expnd. node	Frontier list
	{S}
S	{A,B, C}
A	{B, C, D, E}
B	{C,D,E,G}
C	{D, E, G, F}
D	{E, G, F, H}
E	{G, F, H, G}
G	{F, H, G}



Q2 Text Solution:

The Probability P(Y= 1 | X =1) is given by

$\frac{P(X=1,Y=1)}{P(X=1)}$, First, calculate P(X =1) which is

P(X=1, Y =1) +P(X =1, Y =2)=0.2 + 0.3 = 0.5

Thus, P(Y =1 | X =1) = $\frac{0.2}{0.5} = 0.4$

Q3 Text Solution:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Where: P(A) = Probability of drawing the first marble as red = $\frac{8}{20} = \frac{2}{5}$.

P(B | A) = Probability of drawing the second marble as red given the

So, P (A∩B) = $\frac{2}{5} = \frac{7}{19} = \frac{14}{95}$

Q4 Text Solution:

P(A = 1, B = 0, C = 0, D = 1) = P(A) x P(B|A) x P(C|B) x P(D|B)

= P(A = 1) x p(B = 0|A = 1) x p(C = 0|B = 0) x p(D = 1|B = 0)

= 0.4 x 0.4 x 1 x 0.6 = 0.0%

Q5 Text Solution:

P(Alarm2 | Burglary, Alarml) = P(Alarml, Alarm2, Burglary) / P(Burglary, Alarml) = 0.016/0.072 ~ 0.22

with

P(Alarml, Alarm2, Burglary) = P(Alarml) P(Alarm2) P(Burglary | Alarml. Alarm2) = 0.1 x 0.2 x 0.8 = 0.016

P(Alarml, Ø Alarm2. Burglary) = P(Alarml) P(Ø Alarm2) P(Burglary | Alarml, Ø Alarm2) =0.1 x 0.8 x 0.7 = 0.056

P(Burglary, Alarml) = P(Alarml, Alarm2, Burglary) + P(Alarml, Ø Alarm2, Burglary) = 0.016 + 0.056 = 0.072

Q6 Text Solution:

Each binary variable has 2 possible states. Therefore, the total number of unique joint assignments is 2∧4=16

Q7 Text Solution:

Now, let's examine the truth table for (p ∨ g) → r:

p	q	p∨q	r	(p∨q) → r
T	T	T	T	T
T	T	T	F	F
T	G	T	T	T
T	F	T	F	F
F	T	T	T	T
F	T	T	F	F
F	F	F	T	R
F	F	F	F	T

Q8 Text Solution:

	S	A	C	D	B	E
S	0	0			0	
A	5	5			5	
B	8	8			8	
C	∞	4			4	
F	∞	∞			9	
D	∞	6			6	
E	∞	∞			4	

$S \to B \to E$

Q9 Text Solution:

S®A®D®E®B®F®G3

Q10 Text Solution:

$$P \wedge ((P \wedge Q) \vee (\neg P \wedge \neg Q)) \vee Q$$

$$= (P \wedge P \wedge Q) \vee (P \wedge \neg P \wedge \neg Q) \vee Q$$

$$= (P \wedge Q) \vee Q$$

$$= Q$$

Q11 Text Solution:

$S \to Q \to G2$

Q12 Text Solution:

n = { }

for empty domain $\forall$x is true